Jesús M. Rueda-Becerril, Ph.D.

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Professional Summary

Computational physicist (Ph.D.) with 10+ years writing scientific code, running computational experiments, and validating models against data. Currently evaluates frontier language models on expert-level physics reasoning and builds neural-network surrogates of expensive simulations. Author of open-source scientific libraries (Paramo, Tleco; ApJ 2024) with code review and unit testing, and mentor of graduate students.

Skills

Languages: Python, C/C++, Fortran 95, R, Rust, Octave, SQL, Java, Julia, Shell
Simulation & Modeling: High-fidelity physics simulation (multiscale/multiphysics, kinetic plasma modeling, radiative processes (synchrotron, inverse Compton), radiative transfer, relativistic MHD, hydrodynamics (SPH), stochastic particle dynamics, Lagrangian particle dispersion), numerical PDEs (Fokker–Planck, advection–diffusion), finite difference/volume methods, numerical linear algebra, model validation, physics-informed synthetic data generation, uncertainty quantification, parameter estimation
ML & Data: Scientific ML, language-model evaluation (scientific reasoning), surrogate modeling, Bayesian inference, Gaussian process regression, gradient-based optimization, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, neural networks, SVM, random forest, JAX (introductory coursework)
HPC & GPU: OpenMP, MPI, OpenACC (GPU exploration), SLURM, PBS, HDF5, large-scale parallel computing (Frontera)
Geospatial: QGIS, raster/vector processing, Sentinel-1/2 multispectral data, DEM terrain analysis, terrain and elevation derivatives, geospatial feature engineering for ML
Tools: Git and code review, Docker, Azure, CI/CD (Jenkins, OpenShift), Kafka, Splunk, Jira, QGIS, ParaView, VisIt

Experience

Independent Contractor Jan 2026 – Present
Mercor Remote
Physics subject-matter expert evaluating frontier language models for multiple leading AI labs.

- Review 50+ expert-level physics, hydrodynamics, astrophysics, and cosmology problems for leading AI labs; write original prompts that stump frontier models and co-develop others with fellow writers.
- Diagnose failure modes in model reasoning and write structured feedback to improve scientific accuracy.
- Benchmark model performance and quality-check solutions to complex STEM problems.

Spatial Data Scientist May 2025 – Jan 2026
TealWaters Seattle, WA
Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.

- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.
- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Engineered features from Sentinel-1/2 multispectral imagery and terrain derivatives for ML land-cover and wetland models; built Python tools and QGIS visualizations.
- Presented results to managers and decision-makers, trained 2 colleagues on the WIP tool, and worked with software engineers to move prototype tools into production.

Independent Researcher & Open-Source Developer Jan 2024 – May 2025
Self-directed Seattle, WA
Continued simulation research and open-source development between industry roles.

- Co-developed Tleco, an open-source Rust/Python toolkit that solves the Fokker–Planck equation coupled to radiative transfer to model radiation from relativistic outflows (ApJ 2024) [9].
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library that simulates wind-driven spore dispersal over forested, hilly terrain, parallelized with RcppParallel and OpenMP.
- Used AI coding assistants in the most recent stages of development; changes were accepted only if they passed the unit-test suite and, for some functions, matched reference solutions.
- Invited talk on machine learning in environmental science at the University of Washington (2024).

Software Engineer Apr 2022 – Jan 2024
Paychex Remote
Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.

- Built Java Kafka consumers that moved data between enterprise databases, deployed through Jenkins and OpenShift CI/CD pipelines.
- Created Splunk dashboards tracking the 200–400 users shared between enterprise systems and Oracle EBS, with alerts that flagged production incidents.
- Prepared, validated, and analyzed Oracle EBS data with SQL.
- Wrote full test suites for 5+ Python services in the CI pipeline, keeping production defect rates low.

Postdoctoral Research Associate Feb 2021 – Apr 2022
Rochester Institute of Technology Rochester, NY
HPC simulations of supermassive black hole binaries in an NSF-funded project.

- Led a 7-person team upgrading PatchworkMHD, a C code for magnetohydrodynamic simulations of supermassive black hole binaries.
- Added black hole spin to the simulations with under 5% runtime overhead.
- Ran and benchmarked large production simulations on Frontera (TACC), about 10,000 node-hours, and identified performance bottlenecks in the parallel code.
- Calibrated Paramo parameters against Fermi-LAT gamma-ray spectra with gradient-based optimization, comparing simulation output with data and updating the model automatically; supervised a graduate student on the same problem.
- Co-authored 2 papers from a multi-institution study of binary neutron star mergers [6, 7] and a third on blazar heating and cooling [8].

Postdoctoral Research Fellow Oct 2018 – Nov 2020
Purdue University West Lafayette, IN
Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).

- Designed and wrote Paramo, an open-source Fortran 95 library that solves the Fokker–Planck equation for relativistic particle distributions coupled to radiative transfer; optimized it with OpenMP and tested GPU offload with OpenACC.
- Paramo has been adopted by other research groups, including Combi & Siegel (2023).
- Served as co-I on two NASA Fermi Guest Investigator grant proposals; the funded one ($68K) supported the finding that blazar jets launch with similar energy per baryon, independently of their power [5].
- Added numerical-integration and OpenMP features to Paramo and built Python tools to compute energy losses and spectra of high-energy particles in gamma-ray burst afterglows [4].
- Mentored 3 graduate students (1 M.S., 2 Ph.D.), co-authoring papers with each.
- Wrote Python scripts for statistical modeling of the COVID-19 outbreak in Mexico with a multidisciplinary group, and co-wrote Spanish-language infographics and blog posts against misinformation.

Postdoctoral Research Fellow Jan 2018 – Sep 2018
Universidad Michoacana de San Nicolás de Hidalgo Morelia, Mexico

- Built a Python pipeline that turned GRTrans black hole simulation images into training data for an SVM classifier of black hole spin (grayscale conversion, intensity normalization, spatial feature extraction).
- Released an open-source Python tool for radiative-transfer spectra and light curves.
- Organized an HDF5 workshop to train graduate students in high-volume data storage.

Graduate Research Assistant Oct 2011 – Jul 2017
Universitat de València Valencia, Spain

- Automated runs of the C-SPEV radiative-transfer code with Shell and Python pipelines: 10–60 simulations per project and tens of GB of HDF5 output, with data-quality checks for downstream analysis.
- Studied how magnetization of shocked plasma shapes the radiation spectra of relativistic internal shocks, using the C-SPEV simulation code.
- Analyzed multi-wavelength observations from the NASA Fermi-LAT telescope and the Very Long Baseline Array (VLBA), including cross-matching sources across catalogs by sky coordinates and fitting log-parabola spectra.
- Fit C-SPEV simulation models to telescope data, identifying spectra and light-curve patterns that quantify plasma magnetization in blazars.
- Extended C-SPEV to compute discrete and continuous spectra from particle distributions of arbitrary shape, without affecting simulation runtime.
- Published 2 first-author papers in MNRAS [2, 3].

Education

Ph.D. in Physics Oct 2011 – Jul 2017
Universitat de València Valencia, Spain
Excellent *cum laude*
Advisors: Prof. Miguel A. Aloy and Dr. Petar Mimica
Thesis: Numerical treatment of radiation processes in the internal shocks of magnetized relativistic outflows

M.Sc. in Physics Aug 2009 – Sep 2011
Universidad Michoacana de San Nicolás de Hidalgo Morelia, Mexico
Advisor: Prof. José A. Cervera
Thesis: *Study of TOV stars with the SPH method* (wrote an SPH hydrodynamics code to study the stability of relativistic stars)

B.Sc. in Physics Aug 2004 – Dec 2008
Universidad Autónoma del Estado de México Toluca, Mexico
Advisor: Prof. Francisco S. Guzmán
Thesis: *Numerical solution of null geodesics for the generation of gravitational lenses produced by spherically-symmetric and static spacetimes*
Lic. Juan Josafat Pichardo Cruz Award

Publications

Articles (9 published, 3 first-author; 3 in preparation)

[12] Winkowski, S., **Rueda-Becerril, J. M.**, & Willing, C. *Using Numerical Simulations of Wind on Hilly Terrain to Model Propagule Dispersal*, in prep.

[11] Holzman, L., **Rueda-Becerril, J. M.**, Creamer, C., & Willing, C. *Random Forest Link Landscape Position to Drought Resilience in Giant Sequoia Forests*, in prep.

[10] **Rueda-Becerril, J. M.**, Winkowski, S., & Willing, C. *WindsOfChange: A New Toolkit for Modeling Spore Dispersal in Hilly Terrains*, in prep.

[9] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ 976, 182 (2024), arXiv:2405.17581.

[8] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, MNRAS 513, 5766–5779 (2022), arXiv:2201.07790.

[7] Lopez-Armengol, F. G., Etienne, Z. B., [...], **Rueda-Becerril, J. M.**, [...] *Handing off the outcome of binary neutron star mergers for accurate and long-term postmerger simulations*, Phys. Rev. D 106, 083015 (2022), arXiv:2112.09817.

[6] Murguia-Berthier, A., Noble, S., [...], **Rueda-Becerril, J. M.**, [...] *HARM3D+NUC: A New Method for Simulating the Post-merger Phase of Binary Neutron Star Mergers with GRMHD, Tabulated EOS, and Neutrino Leakage*, ApJ 919, 95 (2021), arXiv:2106.05356.

[5] **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS 501, 4092–4102 (2021), arXiv:2009.02273.

[4] Zhang, H., Christie, I., Petropoulou, M., **Rueda-Becerril, J. M.**, & Giannios, D. *Inverse Compton Signatures of Gamma-Ray Burst Afterglows*, MNRAS 496, 974–986 (2020), arXiv:1910.14049.

[3] **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS 468, 1169–1182 (2017), arXiv:1612.06383.

[2] **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *The influence of the magnetic field on the spectral properties of blazars*, MNRAS 438, 1856–1869 (2014), arXiv:1310.5441.

[1] Guzmán, F. S., & **Rueda-Becerril, J. M.** *Spherical boson stars as black hole mimickers*, Phys. Rev. D 80, 084023 (2009), arXiv:1009.1250.

Conference Proceedings

[P5] **Rueda-Becerril, J. M.** *A numerical approach for radiative cooling in relativistic outflows*, Astron. Nachr. 342, 277–282 (2021), 9th International Workshop on Astronomy and Relativistic Astrophysics, arXiv:2011.13797.

[P4] **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *The blazar sequence revised*, Astron. Nachr. 342, 147–152 (2021), 9th International Workshop on Astronomy and Relativistic Astrophysics, arXiv:2011.13805.

[P3] **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *Numerical simulations of the internal shock model in magnetized relativistic jets of blazars*, PoS(SWIFT 10) 233, 159 (2014), arXiv:1502.07882.

[P2] **Rueda-Becerril, J. M.**, Mimica, P., Aloy, M. A., & Aloy, C. *Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets of blazars*, EPJ Web Conf. 61, 02007 (2013), arXiv:1309.4612.

[P1] Mimica, P., Aloy, M. A., **Rueda-Becerril, J. M.**, Tabik, S., & Aloy, C. *Numerical simulations of dynamics and emission from relativistic astrophysical jets*, J. Phys.: Conf. Ser. 454, 012001 (2013), arXiv:1211.1794.

Grants and Fellowships

Research Grants

NASA Fermi Guest Investigator Program, Cycle 12 2019
A simple model to understand the blazar sequence Grant #121077, $68K
PI: D. Giannios; Co-I: J. M. Rueda-Becerril

Fellowships

International Postdoctoral Fellowship Oct 2018 – Nov 2020
National Council of Science and Technology (CONACyT), Mexico

Postdoctoral Fellowship Jan 2018 – Sep 2018
Secretariat of Public Education, Mexico (Program for the Professional Development of Higher Education Institutions)

Study-Abroad Fellowship Sep 2014 – Aug 2016
National Council of Science and Technology (CONACyT), Mexico

Santiago Grisolía Fellowship Oct 2011 – Jun 2014
Council of Education, Research, Culture and Sport, Valencian Community, Spain

M.Sc. Fellowship Sep 2009 – Aug 2011
CONACyT, at the Institute of Physics and Mathematics, Universidad Michoacana de San Nicolás de Hidalgo

Summer Research Fellowship (XVII Summer of Scientific Research) Jun 2007 – Aug 2007
Mexican Academy of Sciences

Software and Projects

WindsOfChange Aug 2024 – Present
Lagrangian particle-dispersion library (C++/R) github.com/UW-ForestMycobiomeLab/WindSporesHills

- Translated an existing MATLAB code to R and C++, producing a library for Lagrangian particle dispersion over hilly terrain with forest canopy, driven by prescribed wind fields, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004). Developed with UW collaborators; manuscript in preparation.
- Parallelized particle tracking with RcppParallel and OpenMP; designed for scalability to tens of thousands of particles across kilometer-scale domains.
- Used AI coding assistants in the most recent stages of development; changes were accepted only if they passed the unit-test suite and, for some functions, matched reference solutions.

Blazar Population ML Jan 2026 – Present
Physics-informed ML (Python, scikit-learn, PyTorch), developed with Claude github.com/altjerue/ilwikak-ml

- Applying classical ML (PCA, UMAP, Gaussian mixture models, HDBSCAN) to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to test whether physical jet properties can be recovered from multi-wavelength observables; documented missingness and selection effects feature by feature.
- Fitting Gaussian process regression to a lower bound on the jet Lorentz factor for a 334-source MOJAVE cross-matched subsample.
- Building a neural-network surrogate of the forward model (jet parameters to observable SEDs), to be benchmarked against Tleco simulations, to enable fast parameter inference.
- Forecasting blazar gamma-ray flares with a time-series foundation model fine-tuned on Tleco-generated synthetic light curves (in progress).

Tleco Jan 2024 – Sep 2024
Radiative-transfer toolkit (Rust/Python) github.com/zkdavis/Tleco

- Numerical PDE solvers, particle-distribution algorithms, and spectral calculations for relativistic plasmas; successor to Paramo [9].

Paramo Oct 2018 – Apr 2022
Radiative-transfer library (Fortran 95, OpenMP) github.com/altjerue/paramo

- Fokker–Planck and radiative-transfer solver for relativistic particle distributions, parallelized with OpenMP; used by other groups, including Combi & Siegel (2023).

PatchworkMHD Feb 2021 – Apr 2022
Large-scale MHD simulation code (NSF project)

- Led a 7-person upgrade of this production C code for HPC environments, adding black hole spin with under 5% runtime overhead.

Awards

Marcos Moshinsky Award for Best Poster Sep 2020
9th International Workshop on Astronomy and Relativistic Astrophysics (IWARA 2020), video conference

Lic. Juan Josafat Pichardo Cruz Award 2009
Universidad Autónoma del Estado de México
For defending a licentiate thesis within a year of completing the undergraduate coursework.

Presentations

Invited Talks

The Role of Machine Learning in Environmental Science Mar 2024
Seminar on Climate Change Ecology, University of Washington Seattle, WA

Morphology of the spectra from numerical simulations of the internal shocks model for blazars Feb 2019
Astrophysics Seminar, Purdue University West Lafayette, IN

Numerical simulations of the internal shocks model in magnetized relativistic jets of blazars Jun 2018
DATA Group Weekly Seminar, Instituto de Astronomía, UNAM Mexico City, Mexico

Numerical treatment of non-thermal radiation in the internal shocks model for blazars Mar 2018
Weekly Seminar, Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo Morelia, Mexico

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars Oct 2014
IVICFA's Fridays: Computation in Physics, IFIC Paterna, Spain

Contributed Talks

Simulations of supermassive binary black holes accretion dynamics in the spinning case Apr 2022
APS April Meeting 2022 New York, NY

A numerical approach to the Klein-Nishina corrections of radiative cooling in relativistic outflows Apr 2021
APS April Meeting 2021 Online

The blazar sequence revised Sep 2020
9th International Workshop on Astronomy and Relativistic Astrophysics Online
Video: youtube.com/watch?v=BAZNWLNT69M

Influence of the magnetic field on the spectral properties of blazars in the internal shocks scenario Jun 2014
Extreme-Astrophysics in an Ever-Changing Universe Ierápetra, Greece

Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets Jul 2013
XXXIV Biennial Meeting of the Royal Spanish Society of Physics Valencia, Spain

Posters

A numerical approach for radiative cooling in relativistic outflows Sep 2020
9th International Workshop on Astronomy and Relativistic Astrophysics Online
Marcos Moshinsky Award for Best Poster. Video: youtube.com/watch?v=0TJiKg7kOPI

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars Dec 2014
Swift: 10 Years of Discovery Rome, Italy

Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets Jun 2013
The Innermost Regions of Relativistic Jets and Their Magnetic Fields Granada, Spain

Teaching and Mentoring

Thermodynamics (graduate level) Jun 2018
Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo Substitute Instructor

Zachary Davis, graduate student 2018 – 2022
Department of Physics and Astronomy, Purdue University Mentoring
Co-authored papers [8, 9]

Amanda O. Harrison, graduate student 2018 – 2020
Department of Physics and Astronomy, Purdue University Mentoring
Co-authored paper [5]

Hao Zhang, graduate student 2018 – 2019
Department of Physics and Astronomy, Purdue University Mentoring
Co-authored paper [4]

Supervised graduate student (Paramo calibration) 2021 – 2022
Rochester Institute of Technology Mentoring

HDF5 workshop for graduate students 2018
Universidad Michoacana de San Nicolás de Hidalgo Organizer

El origen del universo y el sistema solar (The Origin of the Universe and the Solar System) Feb 2026
Biology students, Facultad de Ciencias, Universidad Autónoma del Estado de México Talk

Professional Development

High Performance Computing on Frontera May – Jun 2021
Jason Allison et al., TACC, Austin, TX Lecture series

Writing Winning Grants Nov 2019
Dr. Lauren Broyles, Purdue University Lecture

XSEDE HPC Workshop: Summer Boot Camp Jun 2019
John Urbanic, Purdue University Workshop

Data Analysis and Machine Learning with Python Feb 2017
Dr. Alejandro Torres, Universitat de València Workshop

Numerical Relativity Simulations of BBH Coalescence Using the Einstein Toolkit Jul 2016
Dr. Vassilios Mewes, Universitat de València Workshop, 8 hours

The Universe in the Light of PLANCK and BICEP2 May 2014
Prof. Nick Mavromatos, Universitat de València Lecture series, 2 credits

Dark Matter Sep 2013
Prof. Alejandro Ibarra, Universitat de València Lecture series, 2 credits

International Cargèse School on Cosmic Accelerators Apr – May 2013
Institut d'Études Scientifiques de Cargèse, France Summer school

Introduction to C++ Programming Apr 2012
Dr. Jacek Generowicz, Universitat de València Workshop, 6 credits

Numerical Relativistic Astrophysics Mar – Apr 2012
Prof. Luciano Rezzolla, Universitat de València Lecture series, 9 hours

Fortran for Scientific Computing Mar 2012
HLRS, University of Stuttgart, Germany Workshop, 33 hours

Outreach

El origen del universo y el sistema solar (The Origin of the Universe and the Solar System) Feb 2026
Biology students, Facultad de Ciencias, Universidad Autónoma del Estado de México Talk

The Role of Machine Learning in Environmental Science Mar 2024
Seminar on Climate Change Ecology, University of Washington Seattle, WA

Los más rápidos y los más furiosos (The Fastest and the Most Furious) Sep 2020
Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco Online talk

Una simulación de la física y la astrofísica (A Simulation of Physics and Astrophysics) Aug 2020
Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco Online talk

ANITA y la teoría de los universos paralelos (ANITA and the Theory of Parallel Universes) May 2020
Científicos Mexicanos en el Extranjero Blog post
mexiciencia.github.io/post/anita

¿Qué es el modelo SIR? (What is the SIR Model?) May 2020
Científicos Mexicanos en el Extranjero Blog post
mexiciencia.github.io/post/modelo-sir

Evolución del brote epidémico de COVID-19 (Evolution of the COVID-19 Epidemic Outbreak) Apr 2020
Científicos Mexicanos en el Extranjero Blog post
Data analysis and modeling collaborator. mexiciencia.github.io/post/covid19

Annual Department of Physics and Astronomy Poster Event Nov 2019
Purdue University 3 posters

Post-Doc Panel Q&A: What Happens When We Complete Our PhDs? Apr 2019
Department of Physics and Astronomy, Purdue University Panelist

Annual Department of Physics and Astronomy Poster Event Nov 2018
Purdue University Poster

¿Decía Einstein la verdad? (Was Einstein Telling the Truth?) Mar 2009
Facultad de Ciencias, Universidad Autónoma del Estado de México Talk

Service

Organizing contributor Dec 2012
X Scientific Meeting of the Spanish Astronomical Society Valencia, Spain

Student representative, Governing Council of the Faculty of Sciences Aug 2007 – May 2009
Universidad Autónoma del Estado de México Toluca, Mexico

Certifications

Gen AI Intensive Kaggle
Generative AI

Data Scientist Professional with Python DataCamp
Python programming, data science, data communication, machine learning

Machine Learning Scientist with Python DataCamp
Machine learning, NLP, deep learning, image processing, big data

Mathematical Foundations of Machine Learning Udemy
Python programming, data science, statistics, machine learning

Python for Statistical Analysis Udemy
Python programming, data science, data communication, machine learning

Languages

Spanish (native; talks and teaching in Spanish), English (full professional proficiency), Catalan, Nahuatl (basic)